thrown in a pond or sound waves coming from two stereo speakers. Wave interference isn't mysterious: in spots where the peaks of one wave line up with the valleys of another, they cancel out and the waves vanish; when the peaks of both waves line up with each other, they're amplified. This creates the patterns of dark and light bands in Figure 5.2.

The weirdness really begins when you shine a much dimmer light on the double slit. Rather than shining a flashlight on the double slit, send the minimum amount of light possible: one photon at a time. Now, each photon faces a choice, like our nanometer Hamlet from the Introduction: go through the left slit or the right slit? Once a photon goes through a slit, it hits the photographic plate behind the slit, leaving a dot on impact. Repeat this over and over again, and you might expect to see two groups of dots, one lined up behind each slit (Figure 5.3a). After all, photons are particles—little tennis balls of light. If you threw tennis balls through a (much larger) double slit, you'd expect them to mostly hit the back wall in two clusters, one behind each slit. But photons aren't really tennis balls of light, and they do something extraordinary instead: though each one hits the plate in a single location, their impacts collectively form an interference pattern on the plate (Figure 5.3b). Even though each photon went through the double slit individually, they still somehow "knew" where to arrive on the photographic plate in order to form an interference pattern. Something was interfering with each photon as it went through, despite the fact that particles don't interfere with each other, and there was only one particle in the double slit at a time anyhow.

Puzzled by the results of your experiment, you repeat it, but with a twist. This time, you attach a little photon detector to each slit, in an attempt to determine which slit each photon goes through, so you can figure out how the interference pattern on the plate is formed. The results convince you of what you had already suspected but hadn't dared to believe: the photons are deliberately messing with you. Now that you're watching them so closely, they refuse to form an interference pattern at all and instead form exactly the two groups of dots that you had expected before (Figure 5.3a). What gives? How can the photons behave differently just because you're watching them? How do they know you're watching them at all?

The Copenhagen interpretation, true to form, gives a mystical pseudo-answer steeped in the language of Bohr's philosophy of complementarity.